	Case Name: Railway Bridge (4)	Sector	**Construction (civil)**	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	■
			Schedule with costs	■
		Risk Analysis	Random simulation	■
			One of nine std. scenarios	
Submitted by	Mieke De Saedeleer		User defined distributions	
Date	2016	Project Control	Automatic tracking	
File Name	C2016-04 Railway Bridge (4)		Tracking based on user input	■

1. Project description

Project authenticity

This project focuses on the construction or renewal of a railway bridge. It starts with planning and site preparation activities, followed by removal or modification of existing structures. The main phase includes building the new bridge elements and installing structural components. The project ends with finishing works, inspections, and preparing the bridge for operational use.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	29
Planned Duration (PD)	248
Budget At Completion (BAC)	906253.867
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	64%
Activity Distribution (AD)	62%
Length of Arcs (LA)	0%
Topological Float (TF)	71%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	14%	16%	2.66
CRI-rho	19%	18%	1.66
CRI-tau	19%	32%	2.13

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	65%	49%	-0.69
SI	69%	43%	-0.74
SSI	10%	17%	3.30
CRI-r	14%	17%	2.73
CRI-rho	15%	16%	2.75
CRI-tau	9%	12%	3.10

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	20%	-18%
PV - SPI	32%	15%
PV - SCI	31%	15%
ED - 1	56%	-3%
ED - SPI	32%	15%
ED - SCI	31%	15%
ES - 1	14%	-12%
ES - SPI(t)	27%	26%
ES - SCI(t)	27%	27%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1%	0%
CPI	1%	0%
SPI	18%	18%
SPI(t)	17%	17%
SCI	18%	18%
SCI(t)	17%	17%
0.8 CPI + 0.2 SPI	12%	12%
0.8 CPI + 0.2 SPI(t)	8%	8%

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 12 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	12061.20	1.89	1	1.01	1.00	1.00
std dev	0	22393.41	3.7	0	0.03	0.06	0.00
final	0	0.00	-4	1	1.00	0.98	1.00

2.3.3.2. Time forecasting

PD	248	Real Duration	242	Late/early	2% early

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	239.17	5.65	1%	1%
PV - SPI	239.67	7.89	2%	1%
PV - SCI	239.67	7.89	2%	1%
ED - 1	239.67	5.61	1%	1%
ED - SPI	240.00	8.08	2%	1%
ED - SCI	240.00	8.08	2%	1%
ES - 1	240.17	3.74	1%	1%
ES - SPI(t)	243.92	15.84	3%	-1%
ES - SCI(t)	243.92	15.84	3%	-1%

2.3.3.3. Cost forecasting

BAC	906253.867	Real Cost	906253.867	Over/under budget	0

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	906253.81	0.02	0%	0%
CPI	906253.81	0.02	0%	0%
SPI	908280.43	13358.93	1%	0%
SPI(t)	919906.74	52365.73	2%	-2%
SCI	908280.43	13358.93	1%	0%
SCI(t)	919906.74	52365.73	2%	-2%
0.8 CPI + 0.2 SPI	906615.84	2609.84	0%	0%
0.8 CPI + 0.2 SPI(t)	908557.12	9068.29	0%	0%